Idea 1:
AI Governance OS & Compliance Agent

Purpose

This document captures two related long-term ideas:
(1) Enterprise AI Governance OS that acts as a runtime control layer for AI systems,

(2) AI Compliance Copilot that automates repetitive work for compliance teams. The goal is to explore engineering problems rather than provide legal advice.

Problem Statement
Enterprises are rapidly deploying AI agents that access internal systems, yet governance is fragmented. Security, compliance, and engineering teams often rely on disconnected tools, spreadsheets, and manual reviews.

Idea 1: Enterprise AI Governance OS
A middleware platform sitting between AI agents and enterprise systems. It logs every AI action, enforces runtime policies, supports human approval workflows, blocks unauthorized access, and produces audit evidence.

How It Works
AI Agent Governance Layer SAP/Microsoft/Databases/APIs. Every request is checked against policies before execution. Sensitive actions can be blocked or paused for approval. All events are logged in a tamper-evident audit trail.

Business Value
• Reduces manual investigations.
• Provides centralized AI governance across multiple vendors.
• Speeds up the process of internal audits.
• Reduces engineering effort to implement controls separately in every application.

Idea 2: AI Compliance Copilot
An AI assistant for compliance teams that collects evidence, summarizes regulations, drafts reports, searches documentation, answers audit questions, and automates repetitive compliance workflows.

Comparison
The **Compliance Copilot** improves human productivity. The Governance OS prevents policy violations at runtime. They can evolve together: start with the Copilot, then add audit logging, then runtime policy enforcement.

Potential Competitors
Governance and compliance: OneTrust, Vanta, TrustArc, BigID.
AI observability: LangSmith, Helicone, Arize AI.
Cloud governance: Microsoft Purview, AWS AI governance capabilities, Google Cloud governance.

Most focus on documentation, discovery, or observability. Few provide vendor-neutral runtime enforcement across multiple AI systems used at an org or a company.

• Machine unlearning orchestration for GDPR erasure requests.
• Vendor-neutral runtime policy enforcement for AI agents.
• Tamper-evident AI audit trails spanning multiple systems.
• Continuous AI risk scoring.
• Human oversight mechanisms for autonomous agents.
• Unified governance across SAP, Microsoft 365, cloud providers, and custom AI.

Phases (Maybe)
Phase 1: AI Compliance Copilot.
Phase 2: AI Audit Gateway.
Phase 3: Runtime Policy Engine.
Phase 4: Enterprise AI Governance Platform.

Questions for Industry Experts
1. Where do existing enterprise security controls fall short for AI agents?
2. Which governance capabilities are customers asking for today?
3. Which component would deliver the highest value as a first open-source module?
4. Which integrations matter most: SAP, Microsoft, cloud providers, or AI frameworks?

RESEARCH TO DO:

Artificial Intelligence is rapidly being adopted across industries, from manufacturing and finance to healthcare and public administration. At the same time, governments are introducing new regulations to ensure that AI systems are safe, transparent, secure, and legally compliant. Organizations must now comply not only with traditional regulations such as GDPR but also with emerging frameworks such as the EU AI Act, NIS2, and industry-specific standards.

Today, compliance is largely a manual process involving legal teams, compliance officers, quality managers, and auditors. They spend significant time reading regulations, interpreting legal requirements, comparing them with internal policies, preparing audit evidence, and tracking regulatory changes.

The objective of this project is to investigate whether AI, document intelligence, and knowledge-based systems can automate these repetitive tasks while keeping humans responsible for final compliance decisions.

Rather than creating another chatbot, the long-term vision is to build an AI-powered Compliance Operating System that continuously assists organizations in understanding regulations, monitoring compliance status, identifying risks, and preparing for audits.

Project Description

The proposed platform aims to become a centralized compliance intelligence system for organizations.

The platform should be capable of:

- Understanding regulations and legal documents.
- Extracting obligations and compliance requirements.
- Comparing regulations with company policies and procedures.
- Detecting compliance gaps.
- Tracking regulatory changes over time.
- Assisting with audit preparation.
- Generating explainable compliance reports.
- Providing continuous compliance monitoring.

The system is intended to support compliance professionals rather than replace them.

Core Problem Statement

Organizations struggle to efficiently manage the increasing number of regulations, internal policies, and compliance requirements.

Current compliance processes are:

- Time-consuming
- Manual
- Expensive
- Difficult to scale
- Highly dependent on human expertise

This creates delays, increases operational costs, and raises the risk of missing important regulatory obligations.

The research question is:

How can AI assist organizations in continuously understanding, monitoring, and managing regulatory compliance while maintaining transparency, explainability, and human oversight?

Research Objectives

The project aims to investigate whether AI can:

- Reduce manual compliance effort.
- Improve regulatory understanding.
- Detects compliance gaps automatically.
- Improve audit preparation.
- Generate explainable compliance recommendations.
- Monitor regulatory changes continuously.
- Assist compliance teams in decision-making.

Individual Problem Statements

Problem Statement 1

Organizations struggle to determine which regulations apply to their business.

Research Goal:
Can AI automatically identify relevant regulations based on industry, geography, and business activities?

Problem Statement 2

Compliance teams manually read hundreds of pages of legal documents.

Research Goal:
Can Large Language Models accurately extract obligations, deadlines, penalties, and responsibilities?

Problem Statement 3

Organizations find it difficult to compare regulations with internal policies.

Research Goal:
Can AI automatically detect missing requirements or conflicting policies?

Problem Statement 4

Regulations change frequently.

Research Goal:
Can AI automatically identify regulatory changes and explain their business impact?

Problem Statement 5

Preparing audit evidence requires collecting documents from multiple departments.

Research Goal:
Can AI organize and retrieve audit evidence automatically?

Problem Statement 6

Compliance risks are usually identified only during audits.

Research Goal:
Can AI predict compliance risks before they become critical?

Problem Statement 7

Compliance knowledge is scattered across emails, policies, contracts, SharePoint, and other systems.

Research Goal:
Can a unified knowledge graph improve compliance intelligence?

Research Questions

The following questions require investigation before building the platform:

Business Questions

- What are the biggest compliance pain points for European companies?
- Which industries spend the most time on compliance?
- Which compliance activities are the most expensive?
- What is the current workflow inside compliance teams?
- How much time is spent preparing for audits?
- Which tasks are repetitive enough to automate?

Technical Questions

- Which AI architecture is most suitable?
- How should legal documents be represented?
- How can retrieval quality be improved?
- How should regulations be versioned?
- How can conflicting regulations be handled?
- How can AI explain every recommendation?
- How can hallucinations be minimized?
- What evaluation metrics should measure compliance accuracy?

Product Questions

- Who should be the first customer?
- Which single feature provides the highest value?
- What is the smallest MVP that companies would pay for?
- Should the platform focus on SMEs or enterprises?
- Which integrations are most valuable (Microsoft 365, SharePoint, SAP, Jira, etc.)?
- How should pricing scale with company size?

Legal & Ethical Questions

- How should confidential documents be protected?
- How should user permissions be managed?
- What level of human review should always remain mandatory?
- What evidence should AI provide for every recommendation?
- How should liability be handled if AI produces an incorrect compliance suggestion?

Expected Long-Term Vision

The long-term goal is to build an Enterprise Compliance Operating System capable of continuously understanding regulations, monitoring organizational compliance, assisting human experts, and providing explainable AI-driven recommendations.

The platform would evolve module by module rather than attempting to solve every compliance problem from the beginning.

Current Status

This project is currently in the **problem validation and research phase**.

The next steps are:

1. Interview compliance professionals.
2. Study existing compliance workflows.
3. Analyze competitor products.
4. Validate the highest-value problem.
5. Design a minimum viable product (MVP).
6. Develop and test the first prototype.